

1. Administrative & Study Identification

Full Study Title: A Phase III, Open-label, Multi-Center, Randomized Study Comparing 177Lu-PSMA-617 vs. a Change of Androgen Receptor-Directed Therapy in the Treatment of Taxane Naïve Men With Progressive Metastatic Castrate Resistant Prostate Cancer **Short Title/Acronym:** PSMAfore **Protocol Number:** CAAA617B12302 **NCT Number:** NCT04689828 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** 177Lu-PSMA-617 (lutetium-177 vipivotide tetraxetan / Pluvicto) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective

Primary Objective: To determine whether 177Lu-PSMA-617 improves Radiographic progression-free survival (rPFS) compared to a change in Androgen Receptor-Directed Therapy (ARDT) in taxane-naïve men with progressive metastatic castrate-resistant prostate cancer (mCRPC). **Secondary Objectives:** To evaluate Overall Survival (OS), Objective Response Rate (ORR), Duration of Response (DOR), Health-related quality of life (FACT-P), and overall safety/tolerability.

2.2 Background & Rationale To address the critical need for effective life-prolonging therapies in patients with PSMA-positive metastatic castration-resistant prostate cancer (mCRPC) who have progressed on a prior androgen receptor pathway inhibitor (ARPI) and are considered appropriate candidates for delaying taxane-based chemotherapy. 177Lu-PSMA-617 is a targeted radioligand therapy that binds specifically to PSMA-expressing cancer cells to deliver localized radiation, providing a crucial alternative to immediate taxane chemotherapy.

3. Study Design & Methodology

3.1 Design Framework

Study Type: Interventional **Control Type:** Active Comparator (Change of ARDT: abiraterone or enzalutamide) **Blinding:** Open-label **Randomization:** 1:1 Randomized (with allowed crossover to 177Lu-PSMA-617 upon confirmed radiographic progression)

3.2 Planned Enrollment & Duration

Target \$N\$: 468 patients **Number of Sites:** 74 sites across Europe and North America (approximately 17 countries) **Estimated Study Start:** 15-Jun-2021 **Estimated Primary Completion:** Follow-up duration of approximately 43 months

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men (\$\ge\$ 18\$ years). 2. Confirmed diagnosis of progressive metastatic castrate-resistant prostate cancer (mCRPC). 3. Taxane-naïve: must not have been exposed to a taxane-containing regimen in the CRPC or mHSPC settings. 4. Progressed on 1 prior ARPI and deemed a candidate for an ARPI change. 5. At least 1 PSMA-positive lesion and no exclusionary PSMA-negative lesions confirmed by 68Ga-PSMA-11 PET/CT.

4.2 Exclusion Criteria 1. Candidates for PARP inhibition therapy. 2. Prior systemic radiotherapy within 12 months. 3. Previous treatment with any taxane-containing chemotherapy regimen for prostate cancer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen

Dosage/Frequency: 177Lu-PSMA-617 administered at a dose of 7.4 GBq (200 mCi) \$\pm\$ 10% once every 6 weeks for up to 6 cycles. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 6 cycles (approximately 36 weeks).

5.2 Concomitant & Prohibited Therapy

Permitted: Best supportive care, including ongoing Androgen Deprivation Therapy (ADT). Patients in the ARDT active comparator arm are permitted to crossover to receive 177Lu-PSMA-617 upon centrally reviewed radiographic progression (rPD). **Prohibited:** Taxane-based chemotherapy, alternative ARPIs (beyond the designated comparator), and systemic radiotherapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Day -28 to 0	Informed Consent, Eligibility Assessment, 68Ga-PSMA-11 PET/CT scan
2	Day 0	Randomization (1:1), First Dose of 177Lu-PSMA-617 or ARDT Change
3	Every 6 Weeks	IV Infusion of 177Lu-PSMA-617 (for up to 6 cycles), Safety Labs, Efficacy Assessment (Radiographic Scans)
4	Up to 43 months	Survival follow-up, Crossover evaluation upon rPD, Final safety and OS monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Radiographic progression-free survival (rPFS), defined as the time from randomization until radiographic progression or death from any cause. **Measurement Method:** Assessed via Blinded Independent Central Review (BICR) of radiographic images in accordance with Prostate Cancer Working Group 3 (PCWG3)-modified RECIST v1.1 Guidelines.

7.2 Secondary & Exploratory Endpoints * **Overall Survival (OS):** Time from randomization to death due to any cause. * **Objective Response Rate (ORR)** and **Duration of Response (DOR).** * **Health-related Quality of Life:** Evaluated using the FACT-P questionnaire.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous monitoring for Grade ≥ 3 AEs (e.g., anemia, dry mouth) and AEs leading to discontinuation. **Serious Adverse Events (SAEs):** Reported within standard 24-hour timelines; incidence actively tracked compared to the ARDT arm. **Data Monitoring Committee (DMC):** Independent External Data Monitoring Committee to oversee safety and interim OS analysis boundaries.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intention-to-Treat (ITT) for primary efficacy endpoints; Safety Set for AE evaluations. **Sample Size Justification:** Designed for primary analysis at ~ 156 rPFS events and second interim OS analysis at ~ 125 deaths, powered appropriately (overall $\alpha = 0.025$, one-sided). **Handling of Missing Data:** Prespecified crossover-adjusted analysis for OS utilizing rank-preserving structural failure time (RPSFT) methodology to account for the high crossover rate (84.2%) from the ARDT arm upon progression.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring per Novartis oncology trial standards. **Audit Trail:** Robust tracking of Blinded Independent Central Review (BICR) image evaluations, eCRF locking, and rigorous data clarification processes.

11. Study Identifiers & Governance

Primary ID: CAAA617B12302 **Registry Number:** NCT04689828 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** PSMAfore **Official Regulatory Title:** A Phase III, Open-label, Multi-Center, Randomized Study Comparing 177Lu-PSMA-617 vs. a Change of Androgen Receptor-Directed Therapy in the Treatment of Taxane Naïve Men With Progressive Metastatic Castrate Resistant Prostate Cancer

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Progression on 1 prior ARPI and PSMA-positivity **Medical Methodology:** Targeted Radioligand Therapy vs. Androgen Receptor-Directed Therapy (ARDT) **Optimization Target:** Radiographic Progression-Free Survival (rPFS) prolongation and delay of taxane chemotherapy

13. Study Procedures & Methodology

Management Pathway: Targeted therapeutic pathway for taxane-naïve mCRPC **Education Protocol (HCP):** Radioligand therapy administration, safe handling of radioactive isotopes, and PET/CT screening interpretation **Education Protocol (Patient):** Radiation safety precautions following treatment cycles and management of xerostomia (dry mouth) **Quality Audit Mechanism:** Centralized RECIST v1.1 and PCWG3 radiographic image auditing via BICR

14. Observation & Follow-Up Schedule

Total Duration: Approximately 43 months **Onsite Visit Cadence:** Every 6 weeks during the treatment phase (up to 6 cycles) **Remote Monitoring Frequency:** Periodic OS and safety tracking after treatment discontinuation **Checklist Review Interval:** Routine imaging scans evaluated for rPD criteria periodically

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				